

WORKSHEET - NAPLAN Year 9

Number and Algebra
Ratios, Scale and Proportion

Teacher: Alex Latham

Exam Equivalent Time: 11.25 minutes (based on NAPLAN allocation of
~1.25 min/mark)



Worked Solutions

1. Number and Algebra, NAP-02654

$$\begin{aligned}\text{Acres} &= \frac{16}{4} \\ &= 4\end{aligned}$$

2. Number and Algebra, NAP-61529

Ratio 3:2

Let x = Number of Carly's dolls

$$\begin{aligned}\frac{x}{12} &= \frac{2}{3} \\ x &= \frac{2 \times 12}{3} \\ &= 8 \text{ dolls}\end{aligned}$$

3. Number and Algebra, NAP-79138

$$\begin{aligned}\text{Ratio} &= 40:600 \\ &= 1:15\end{aligned}$$

4. Number and Algebra, NAP-02685

$$\begin{aligned}\text{Ratio} &= 3:4 \\ &= 1.5:2\end{aligned}$$

$\therefore$ 1.5 cups of sugar, 2 cups of sultanas is
in the correct ratio.

5. Number and Algebra, NAP-85080

$$\begin{aligned}\text{Ratio is } &1:400 \\ \text{Since } 120 \text{ m} &= 12\,000 \text{ cm,} \\ \text{Length of the scale diagram} \\ &= 12\,000 \div 400 \\ &= 30 \text{ cm}\end{aligned}$$

6. Number and Algebra, NAP-79194

Females : Males = 13 : 11

∴ In the room, there are

26 females and 22 males.

∴ 4 females need to leave for the

ratio to become 1 : 1

7. Number and Algebra, NAP-32158

Ratio is 4 : 6

$$\frac{1}{10} \times 360 = 36$$

⇒ Ratio can also be expressed:

$$4 \times 36 : 6 \times 36$$

$$144 : 216$$

∴ 144 boys attend the school.

8. Number and Algebra, NAP-79201

Original ratio 3 : 4

12 boys : 16 girls

After students leave, new ratio is:

$$10 : 15$$

$$2 : 3$$

9. Number and Algebra, NAP-14576

$$\text{Litres of milk} = \frac{\text{parts milk}}{\text{total parts}} \times 7.0$$

$$= \frac{4}{14} \times 7.0$$

$$= 2.0 \text{ litres}$$